

OpenStack based hands-on experimentation infrastructure

Length	Module
Collection	NSA NCAE-C Initiative
Updated	Invalid Date
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Academic Levels	Undergraduate, Graduate
Link	https://clark.center/details/jianlipan/65c035c6-8b70-499d-8a21-293d018205cb

Description

This module is to provide students with a general overview of the OpenStack based edge cloud infrastructure. It will cover the basic introduction of OpenStack, its key components, requirements, configuration and operations. A typical edge cloud lab infrastructure implementation of OpenStack will be introduced. The detailed installation and configuration of Compute Node, Controller Node, and Storage Node will be discussed. The student will have the opportunity to use and operate the basic OpenStack cloud through exercises. The content of this module is mostly comprehensive surveying and introductory, but also include simple hands-on operation.

Notes

Outcomes

- Practice how to operate and manage OpenStack; and how to launch and configure a Virtual Machine
- Illustrate the structure of the OpenStack edge cloud lab infrastructure and configuration
- Describe the basics of OpenStack cloud platform and its framework
- Experiment With how to install and configure the key OpenStack nodes such as Compute Node, Controller Node, and Storage Node.
- Explain why OpenStack is suitable for edge cloud usage
- Inspect the overall OpenStack experiment deployment

Links

External links that are associated with this learning object

- [OpenStack Website](#)
- [OpenStack Newton](#)

- [OpenStack Image Guide](#)